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Project Phase 1 - Review 2

Deepfake Detection in Low-Resolution KYC Videos Using rPPG and Hybrid Quantum Machine Learning

Presented by

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INTRODUCTION

- **Deepfakes:** AI-generated manipulation of **identity, expressions, voice**
- **Risk:** Major threat to **banking & KYC** (+21% attacks, ~\$900M losses)
- **Challenge:** **Low-resolution video** → weakens traditional detection
- **Solution:** **rPPG-based physiological signals + hybrid models**
- **Key Insight:** Visual artifacts degrade in low resolution, but physiological signals **remain** consistent



Image Source: Gemini



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PROBLEM STATEMENT

- **Input:** Low-resolution live facial video stream
- **Output:** Real–deepfake authentication score
- **Motivation:** To enable secure KYC verification by preventing fraud and reducing false rejections in poor-quality video.
- **Approach:** rPPG-based liveness analysis with hybrid quantum-classical classification



Image Source: Gemini



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OBJECTIVES

- To **extract video frames** and perform **basic standardization** for reliable model input.
- To **analyze frame sequences** and **extract facial cues** using tracking and consistency checks.
- To incorporate **rPPG based liveness detection** for **physiological authentication**.
- To integrate a **quantum-hybrid decision layer** for **improved classification accuracy** and **authentication output**.



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LITERATURE SURVEY

Sl. No.	Title	Year	Methodology	Research gap
1	Short And Low Resolution Deepfake Video Detection Using CNN (2022 IEEE 10th Region 10 Humanitarian Technology Conference (R10-HTC))	2022	• dlib → face crop • InceptionResNetV2 • MobileNet → transfer learning • GAP + Softmax → classification	• Frame-based only • No temporal flow • No rPPG or biological verification
2	Design and analytical performance of a hybrid model using deep learning techniques for the detection of deepfake videos(2024 International Conference on Artificial Intelligence and Quantum Computation-Based Sensor Application (ICAIQSA))	2024	• EfficientNetB1, InceptionV3, Xception → spatial features • GRU → temporal learning • Frame sampling + augmentation → preprocessing • Binary classifier → output	• Visual-pattern based only • No physiological verification • No quantum enhancement or confidence score



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LITERATURE SURVEY

Sl. No.	Title	Year	Methodology	Research gap
3	Quantum-Enhanced Deepfake Detection: A Hybrid Classical-Quantum Deep Learning Approach (2025 IEEE 2nd International Conference on Green Industrial Electronics and Sustainable Technologies (GUEST))	2025	• Xception → local features • ViT → global context • Fusion/compression → QNN input • QNN → final classification	• Still-image based only • No temporal video cues • No low-resolution KYC focus
4	Utilizing rPPG Signal Synchronization and Deep Learning Techniques for Deepfake Video Detection	2025	• Facial rPPG extraction → cheek-based signals • SNR, PSD, MAD, SD, Pearson correlation → signal comparison • CNN-LSTM + attention → classification	• Sensitive to lighting and motion blur • Facial occlusion affects performance • Compression reduces rPPG reliability



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LITERATURE SURVEY

Sl. No.	Title	Year	Methodology	Research gap
5	A Comprehensive Review of Deepfake Detection Techniques Utilizing Remote Photoplethysmography	2025	• rPPG analysis → RGB pulse changes • Facial regions → cheeks and forehead • Biological rhythm comparison → authenticity check	• Not focused on low-resolution financial videos • No quantum enhancement • Lacks uncertainty-based verification
6	Multimodal DeepFake Detection via Audio-Visual Feature Fusion and Temporal Attention (2025 6th International Conference on Big Data, Artificial Intelligence and Internet of Things Engineering (ICBAIE))	2025	• ViT → visual features • Temporal attention → frame dependency • Mel-spectrogram encoder → audio features • Fusion → multimodal classification	• Depends on external audio-visual cues • Ignores internal rPPG signals • Heavy for low-res KYC videos



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LITERATURE SURVEY

Sl. No.	Title	Year	Methodology	Research gap
7	A Quantum-Hybrid Framework for Enhanced Deepfake Detection: QAOA Feature Selection & Quantum-Inspired Attention	2026	• InceptionResnetV1 → high-dimensional features • QAOA → feature selection • Quantum attention → forgery emphasis • Grad-CAM → interpretability	• High compute overhead • Weak cross-dataset robustness • Not tested on financial/KYC
8	Multimodal Vision Transformer Forensics for Deepfake Resistant Secure Authentication (2026 8th International Conference on Intelligent Sustainable Systems (ICISS))	2026	• EfficientNetB1, InceptionV3, Xception → spatial features • GRU → temporal learning • Frame sampling + augmentation → preprocessing • Binary classifier → output	• Visual-pattern based only • No physiological verification • No quantum enhancement or confidence score

METHODOLOGY

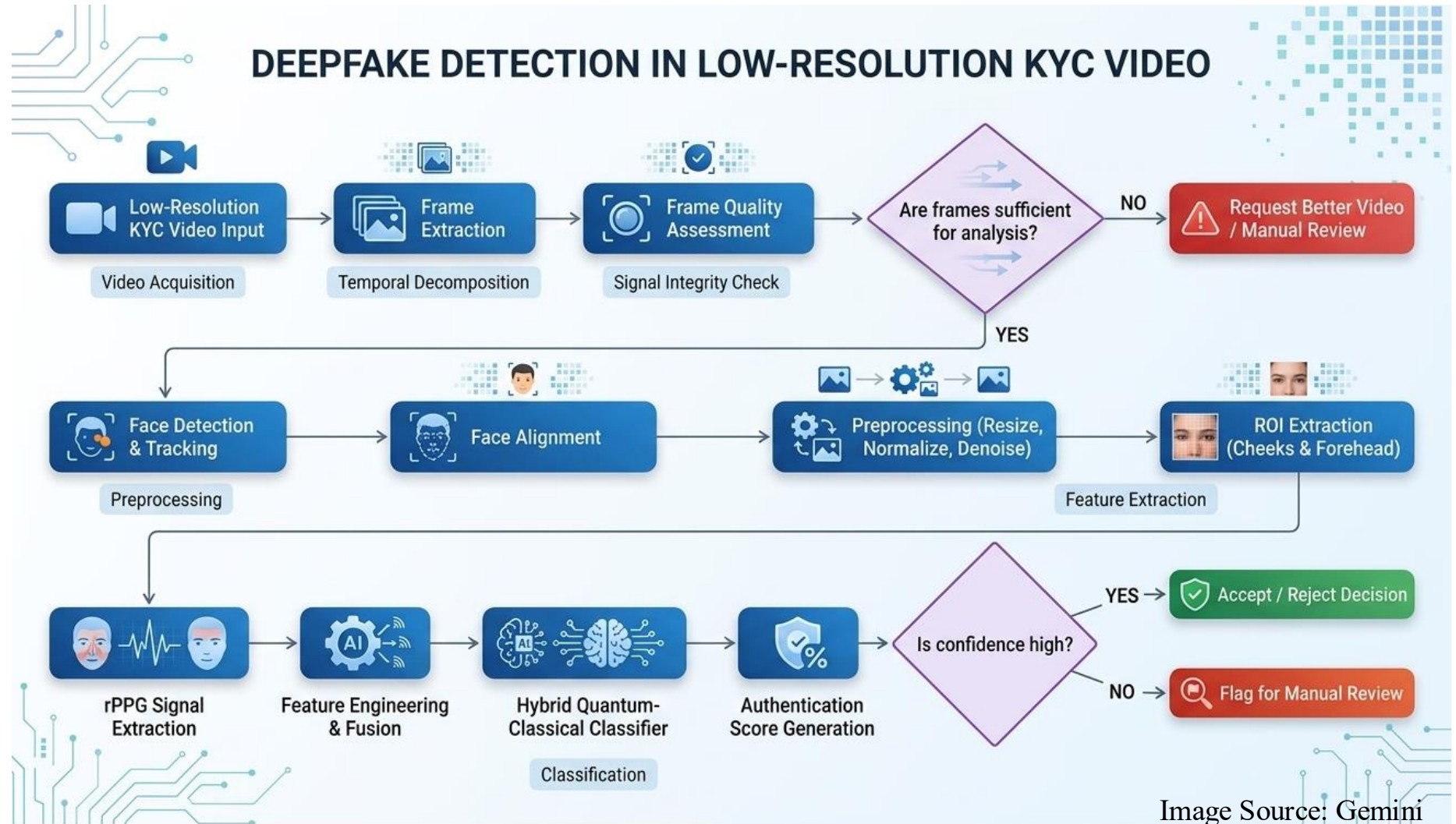


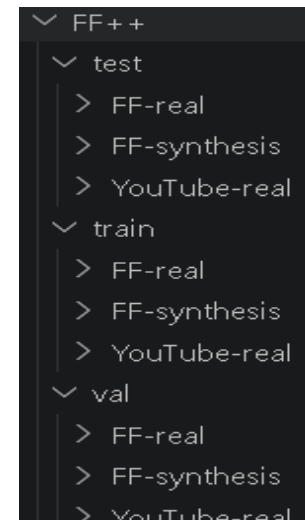
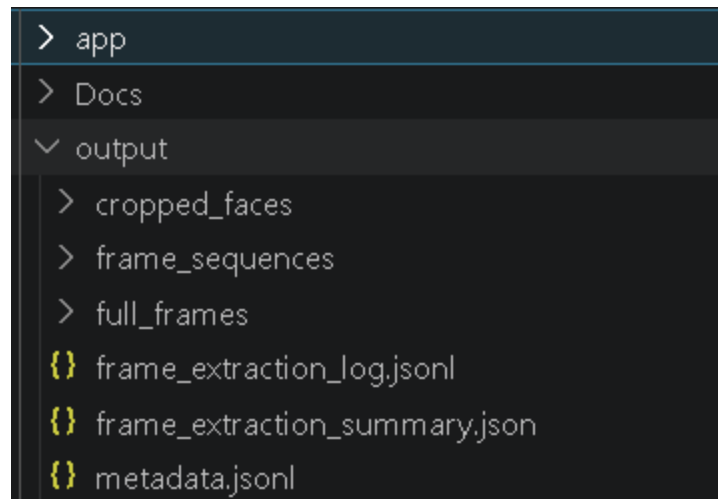
Fig 1: Advanced Deepfake Analysis Methodology for Low-Resolution Video Verification



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OUTCOMES OF PROJECT PHASE 1

- **Dataset Collection:** Collected and organized the dataset required for initial experimentation.
- **Literature Survey:** Reviewed research papers relevant to the proposed system.
- **Methodology Reference:** Identified one key paper to guide the methodological direction of the work.
- **Objective 1 Completion:** Completed Objective 1, which focused on facebased input preparation and early-stage processing design.
- **Review Paper Drafting:** Prepared the review paper based on the studied literature.



→ Dataset: face forencis ++



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FUTURE WORKS

- **Frame Analysis:** Identify stable facial cues from aligned frame sequences
- **rPPG Validation:** Evaluate physiological signals for liveness detection
- **Hybrid Classification:** Implement and test quantum-classical decision model
- **System Evaluation:** Assess performance under varied input condition
- **Research Output:** Document methodology and experimental results

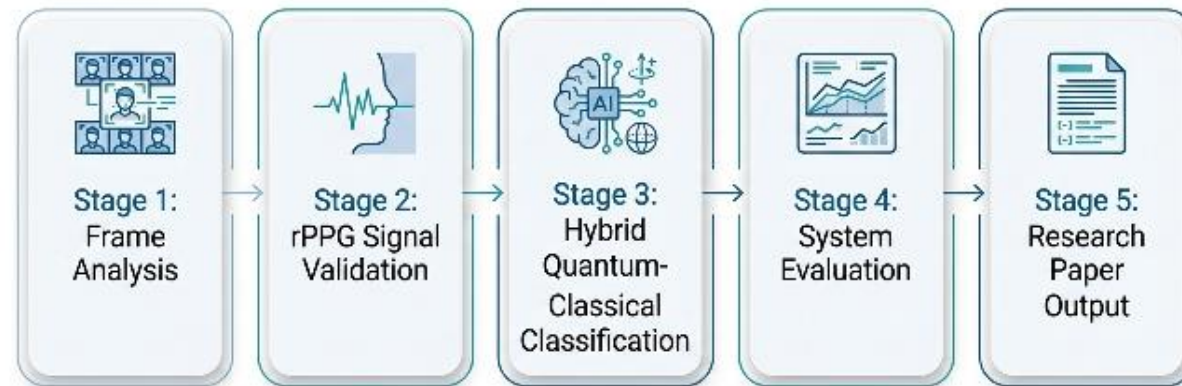


Image Source: Gemini

Fig 2: Future Work diagram



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CONCLUSION



Image Source: Gemini

- **Project Overview:** Deepfake detection for low-resolution KYC video systems
- **Literature Survey:** Analysis of existing deepfake detection approaches
- **Dataset Study:** Identification and organization of relevant datasets
- **Methodology in Progress:** Input preprocessing, rPPG analysis, and model development underway
- **Outcome:** A physiologically grounded approach for reliable deepfake detection in low-resolution environments

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Thank You

Questions?



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